AS

ECONOMICS

7135/1

Paper 1 The Operation of Markets and Market Failure

Practice Paper — Hard

Mark scheme

Mark schemes are working documents. Indicative content is provided as a guide; examiners must credit other valid points. Students do not have to cover all of the indicative content to reach the highest level of the mark scheme. An answer which contains nothing of relevance to the question must be awarded no marks.

SECTION A

The following list indicates the correct answers used in marking students' responses.

Q	Ans	Description	Q	Ans	Description
1	B	is a prediction or factual claim that can in principle be tested against evidence.	11	B	200 (where MSC = MPC + MEC = £16 equals MSB = MB = £16).
2	B	experiencing some unemployment of factors of production.	12	C	a rise in price by less than £2, with the burden shared between producers and consumers in proportions that depend on PED and PES.
3	C	Total revenue is unchanged and total profit could either rise or fall depending on costs.	13	B	£6 [AVC = (TC – TFC)/Q = (900 – 300)/100 = £6].
4	B	–2.5, suggesting the goods are close complements.	14	B	Greater spare capacity and the ability to install new capital equipment.
5	B	A persistent shortage and a possible black market for the foodstuff.	15	B	one firm can supply the entire market at a lower long-run average cost than two or more firms could.
6	B	Quantity supplied is fixed regardless of price changes.	16	B	encourage the firm to lower its costs by cutting prices in real terms each year.
7	A	rival in consumption but non-excludable.	17	B	Present bias — consumers value present consumption disproportionately over future consumption.
8	B	adverse selection, with high-risk individuals being more likely to buy insurance.	18	B	The demand for car factory workers depends on the demand for cars.
9	C	average variable cost.	19	B	the cost of intervention exceeds the benefits, and unintended consequences make resource allocation worse.
10	C	produces a lower output and charges a higher price, generating a deadweight welfare loss.	20	A	firms produce at the lowest point of their long-run average cost curves.

Q4 working: XED = $\%\Delta Q \div \%\Delta P_{related}$ = (+25%) ÷ (–10%) = –2.5. A negative XED indicates complements; |XED| > 1 indicates strong complements.

Q11 working: Social optimum is where MSB = MSC. MSC = MPC + MEC. At Q = 100: MSC = 10, MSB = 20 (under-produced). At Q = 200: MSC = 16, MSB = 16 (optimum). At Q = 300: MSC = 22, MSB = 12 (over-produced). At Q = 400: MSC = 28, MSB = 10 (over-produced).

Q13 working: Total variable cost = TC – TFC = 900 – 300 = £600. AVC = TVC ÷ Q = 600 ÷ 100 = £6.

Levels of response — 25 mark questions

Level	Response	Marks
5	Sound, focused analysis and well-supported evaluation that: • is well organised, showing sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors • includes good application of relevant economic principles to the given context and, where appropriate, good use of data to support the response • includes well-focused analysis with clear, logical chains of reasoning • includes supported evaluation throughout the response and in a final conclusion.	21–25
4	Sound, focused analysis and some supported evaluation that: • is well organised, showing sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors • includes some good application of relevant economic principles to the given context and, where appropriate, some good use of data to support the response • includes some well-focused analysis with clear, logical chains of reasoning • includes some reasonable, supported evaluation.	16–20
3	Some reasonable analysis but generally unsupported evaluation that: • focuses on issues that are relevant to the question, showing satisfactory knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present • includes reasonable application of relevant economic principles to the given context and, where appropriate, some use of data to support the response • includes some reasonable analysis but which might not be adequately developed or becomes confused in places • includes fairly superficial evaluation; there is likely to be some attempt to make relevant judgments but these are not well-supported by arguments and/or data.	11–15
2	A fairly weak response with some understanding that: • includes some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely • includes some limited application of relevant economic principles to the given context and/or data to the question • includes some limited analysis but it may lack focus and/or become confused • includes attempted evaluation which is weak and unsupported.	6–10
1	A very weak response that: • includes little relevant knowledge and understanding of economic terminology, concepts and principles • includes application to the given context which, at best, is very weak • includes attempted analysis which is weak and unsupported.	1–5

Context 1 — The UK fishing industry

2 1 Define 'common access resource'.

[3 marks]

Level	Response	Max 3 marks
3	A full and precise definition is given.	3 marks
2	The substantive content of the definition is correct, but there may be some imprecision or inaccuracy.	2 marks
1	Some fragmented points are made.	1 mark

Examples of acceptable definitions worth 3 marks:

• a resource that is rival in consumption (use by one reduces availability for others) but non-excludable (it is impossible or very costly to prevent others using it)
• a finite resource which no one owns and from which no user can be excluded, so that each user's consumption reduces what is available to all other users

Examples of a definition worth 2 marks:

• a resource that is non-excludable and rival, but without specifying both features clearly
• a resource that anyone can use freely

Examples of a definition worth 1 mark:

• an example of a common access resource (e.g. fish stocks, the atmosphere, common land)
• a free resource

MAXIMUM FOR QUESTION 21: 3 MARKS

2 2 Percentage change in the total wholesale value of UK fish landings, 2014 → 2023.

[4 marks]

Calculation:
Value in 2014 = 711 × 1 480 = £1 052 280 thousand
Value in 2023 = 547 × 2 350 = £1 285 450 thousand
% change = [(1 285 450 – 1 052 280) ÷ 1 052 280] × 100 = 22.158… = **+22.2% (to 1 dp)**.

Response	Max 4 marks
For the correct answer: +22.2% (or a rise of 22.2%)	4 marks
For the correct value, not rounded to one decimal place: e.g. 22.16% OR for the correct value, rounded the wrong way: 22.1% OR for the correct value but with missing/incorrect units (e.g. £22.2 or 22.2 tonnes)	3 marks
For the correct method/figures throughout but the wrong final answer (e.g. arithmetic slip giving 22.3% or 22.0%) OR a response with two of: missing/incorrect units, incorrect rounding, missing/incorrect sign	2 marks
For the correct method (change in total value ÷ original total value × 100) OR a correct intermediate value such as 1 052 280 or 1 285 450 or 233 170 (the absolute change in £ thousand)	1 mark

MAXIMUM FOR QUESTION 22: 4 MARKS

2 3 Two significant features of UK fish landings.

*[4 marks] Award up to 2 marks for **each** significant feature.*

Response	Max 4 marks
Identifies a significant feature Makes accurate use of the data to support the feature identified Unit of measurement given accurately	2 marks
Identifies a significant feature but only one piece of data is given when two are needed and/or no unit of measurement is given and/or the unit of measurement is inaccurate and/or the wrong date is given	1 mark

If a candidate identifies more than two features, reward the best two.

Significant features include:

• Landings are highest in 2017 at 724 thousand tonnes
• Landings are lowest in 2023 at 547 thousand tonnes
• The range is 177 thousand tonnes (547 to 724 thousand tonnes)
• Landings are lower at the end of the period than at the beginning, falling from 711 thousand tonnes in 2014 to 547 thousand tonnes in 2023 — a fall of 164 thousand tonnes (about 23%)
• After a small peak in 2017, landings fall in every subsequent year of the period shown
• The largest single-year fall is between 2018 and 2019, when landings fell from 692 to 622 thousand tonnes — a fall of 70 thousand tonnes
• The smallest single-year change is between 2014 and 2015, when landings fell from 711 to 707 thousand tonnes — a fall of 4 thousand tonnes

Note: Allow a margin of ± 5 thousand tonnes.

MAXIMUM FOR QUESTION 23: 4 MARKS

2 4 PPF diagram — fall in regenerative capacity of wild fish, with fish-farming productivity unchanged.

[4 marks]

Expected diagram: an initial PPC bowed out from the origin (or a straight line) between two appropriately-labelled axes ('Wild-caught fish' and 'Farmed fish'). A second PPC should pivot inward along the wild-caught fish axis only, while still touching the same point on the farmed fish axis. Curves clearly labelled and/or arrows used to indicate the direction of the shift.

Response	Max 4 marks
Accurately drawn initial PPC and a second PPC pivoting inward along the wild-caught fish axis only, with appropriately-labelled axes and a clear indication of which curve is which	4 marks
Accurately drawn initial PPC and a pivoted second PPC but no indication of which is which OR PPC with clear indication of pivot but with at least one axis label missing or inappropriate	3 marks
Accurately drawn initial PPC with labelled axes but no, or an incorrect, second curve (e.g. a parallel inward shift)	2 marks
Accurately drawn initial PPC with at least one axis label missing or inappropriate and no, or an incorrect, second curve OR appropriately labelled axes but with incorrect or missing curve(s)	1 mark

Notes: A title is not required. The exact shape does not matter, but if the PPC does not touch both axes deduct 1 mark. The only appropriate axes labels are 'Wild-caught fish' and 'Farmed fish'.

MAXIMUM FOR QUESTION 24: 4 MARKS

2 5 Explain how the operation of the free market may lead to the over-exploitation of UK fish stocks.

[10 marks]

Level	An answer that:	Marks
3	• is well organised and develops one or more of the key issues that are relevant to the question • shows sound knowledge and understanding of relevant economic terminology, concepts and principles • includes good application of relevant economic principles and/or good use of data to support the response • includes well-focused analysis with a clear, logical chain of reasoning • may include a relevant diagram to support their explanation.	8–10
2	• includes one or more issues that are relevant to the question • shows reasonable knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present • includes reasonable application of relevant economic principles and/or data to the question • includes some reasonable analysis but it might not be adequately developed and may be confused in places • may include a relevant diagram to support their explanation.	4–7
1	• is very brief and/or lacks coherence • shows some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely • demonstrates very limited ability to apply relevant economic principles and/or data to the question • may include some very limited analysis but the analysis lacks focus and/or becomes confused • may include a diagram but the diagram is likely to be inaccurate in some respects or is inappropriate.	1–3

Relevant issues include:

• Definition of a common access resource — rival but non-excludable — applied to wild fish stocks
• The 'tragedy of the commons': each fisher gains the full private benefit of an extra catch but bears only a small share of the wider cost of stock depletion
• Private cost vs social cost — fishing imposes a negative externality on other current and future users of the resource because MSC > MPC
• Diagram of a negative externality of production showing MPC, MSC and the welfare loss from over-production at the free-market equilibrium relative to the social optimum
• Absence of property rights — no one owns the fish until they are caught, removing the incentive to conserve stocks
• Imperfect/asymmetric information — fishers may not know the true stock level; firms may discount the future heavily
• Free-rider problem — if some fishers restrain themselves, others can take the fish they leave behind
• Implication: in the absence of regulation, output exceeds the socially optimal level and stocks may fall below the level needed for population regeneration
• Possible mention of how PED of fish, fuel costs and fleet size influence the rate of depletion

MAXIMUM FOR QUESTION 25: 10 MARKS

2 6 Discuss whether the UK government should impose tighter regulation of UK fishing.

[25 marks]

Areas for discussion include:

• Recent trends — declining UK landings (Extract A); 36% of stocks fished unsustainably (Extract B)
• Wild fish stocks as a common access resource and the case for intervention to correct market failure
• Negative externalities of over-fishing — on future fishers, on marine ecosystems, on coastal communities
• Diagram of a negative externality of production showing MPC, MSC, MSB and the welfare loss; how

regulation can move output toward the socially optimal level

• Methods of tighter regulation — lower total allowable catches (TACs), individual fishing quotas, no-take marine protected areas (MPAs), gear restrictions, seasonal closures, tradable fishing permits, harsher penalties for breaches

• Comparison of regulation vs market-based solutions (tradable permits) — tradable permits give the flexibility to allocate quota to the lowest-cost fishers and can be reduced over time

• Impact on different agents — small-scale fishers (compliance costs already high — Extract B), large industrial trawlers (economies of scale), processors, coastal communities with few alternative jobs (Extract C), consumers (Extract C suggests prices could rise and worsen food inflation), the marine environment, future generations

• Financial cost and opportunity cost — Marine Management Organisation already employs ~240 enforcement officers (Extract B); further enforcement implies higher taxpayer cost

• Risks of government failure — quotas set wrongly, regulatory capture by larger firms, complexity of rules, discards, illegal landings, administrative costs

• 'Leakage' — Extract C suggests tighter UK rules may transfer activity to foreign fleets in adjacent waters, with little or no overall benefit to stocks

• Equity issues — small-scale coastal fishers vs large industrial vessels; jobs vs environment

• Case for leaving more to the existing system — declining fleet size and rising fuel costs (Extract B) may already be reducing pressure on stocks

• Market failure versus government failure

• An overall, supported judgement

The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the candidate's response.

Use the levels mark scheme on the earlier page to award marks for this question.

MAXIMUM FOR QUESTION 26: 25 MARKS

Context 2 — The market for rail travel in Great Britain

2 7 Define 'natural monopoly'.

[3 marks]

Level	Response	Max 3 marks
3	A full and precise definition is given.	3 marks
2	The substantive content of the definition is correct, but there may be some imprecision or inaccuracy.	2 marks
1	Some fragmented points are made.	1 mark

Examples of acceptable definitions worth 3 marks:

• an industry in which a single firm can supply the whole market at a lower long-run average cost than two or more firms could
• a market in which long-run average cost falls continuously over the relevant range of output, so one firm is the most cost-efficient supplier

Examples of a definition worth 2 marks:

• a market with very high fixed costs where one firm is the lowest-cost supplier
• an industry where one firm dominates because of falling average costs

Examples of a definition worth 1 mark:

• a single seller in a market
• example of a natural monopoly (e.g. railway track, water pipes, electricity grid)

MAXIMUM FOR QUESTION 27: 3 MARKS

2 8 Percentage change in GB rail passenger journeys, 2019/20 → 2020/21.

[4 marks]

Calculation: [(388 – 1 739) ÷ 1 739] × 100 = –77.6883… = **–77.7% (to 1 dp)**.

Response	Max 4 marks
For the correct answer: –77.7% (or a fall of 77.7%)	4 marks
For the correct value without the minus sign: 77.7% OR for the correct value, not rounded to one decimal place: e.g. –77.69% OR for the correct value, rounded the wrong way: –77.6% OR for the correct value but with missing/incorrect units: e.g. –£77.7	3 marks
For the correct calculation/figures throughout but the wrong answer OR a response with two of: missing/incorrect units, incorrect rounding, missing sign	2 marks
For the correct method (change ÷ original × 100) OR a correct intermediate value such as –1 351 (the absolute change in millions)	1 mark

MAXIMUM FOR QUESTION 28: 4 MARKS

2 9 Two significant features of GB rail passenger journeys.

*[4 marks] Award up to 2 marks for **each** significant feature.*

Response	Max 4 marks
Identifies a significant feature Makes accurate use of the data to support the feature identified Unit of measurement given accurately	2 marks

Response	Max 4 marks
Identifies a significant feature but only one piece of data is given when two are needed and/or no unit of measurement is given and/or the unit of measurement is inaccurate and/or the wrong date is given	1 mark

Significant features include:

• Journeys are highest in 2018/19 at 1 755 million
• Journeys are lowest in 2020/21 at 388 million, reflecting the impact of the pandemic
• The range is 1 367 million (388 to 1 755 million)
• Journeys were broadly stable between about 1 650 and 1 755 million in the seven years to 2019/20, before the pandemic-related collapse
• The greatest change is between 2019/20 and 2020/21, when journeys fell from 1 739 million to 388 million — a fall of 1 351 million (about 77.7%)
• Journeys at the end of the period (1 385 million in 2022/23) remain below pre-pandemic levels — a fall of 354 million (about 20%) compared with 2018/19
• Journeys recover strongly from 2020/21 to 2022/23, rising by 997 million (from 388 to 1 385 million)

Note: Allow a margin of ± 10 million.

MAXIMUM FOR QUESTION 29: 4 MARKS

3 0 PPF diagram — transfer of resources from road transport to rail transport, with resources remaining fully employed.

[4 marks]

Expected diagram: an initial PPC bowed out from the origin (or a straight line) between two appropriately-labelled axes ('Rail transport' and 'Road transport'), with a movement along the PPC towards the 'Rail transport' axis. This requires an arrow or two labelled points (e.g. X and Y) to indicate the direction of the move.

Response	Max 4 marks
Accurately drawn PPC with appropriately-labelled axes and two points on the curve showing more resources used for rail transport	4 marks
Accurately drawn PPC with appropriately-labelled axes and two points on the curve but no indication of which point is which OR PPC with two points showing the movement, but with at least one axis label missing or inappropriate	3 marks
Accurately drawn PPC with labelled axes and an initial point on the curve but no, or an incorrect, second point OR PPC with labelled axes showing the movement, but with one or both points not on the curve	2 marks
Accurately drawn PPC with at least one axis label missing or inappropriate and no, or an incorrect, second point OR labelled axes but with incorrect or missing curve	1 mark

Notes: A title is not required. If the PPC does not touch both axes, deduct 1 mark. The only appropriate axis labels are 'Rail transport' and 'Road transport'.

MAXIMUM FOR QUESTION 30: 4 MARKS

3 1 Explain factors that could cause rail fares to rise.

[10 marks]

Level	An answer that:	Marks
3	• is well organised and develops one or more of the key issues that are relevant to the question • shows sound knowledge and understanding of relevant economic terminology, concepts and principles • includes good application of relevant economic principles and/or good use of data to support the response • includes well-focused analysis with a clear, logical chain of reasoning • may include a relevant diagram to support their explanation.	8–10
2	• includes one or more issues that are relevant to the question • shows reasonable knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present • includes reasonable application of relevant economic principles and/or data to the question • includes some reasonable analysis but it might not be adequately developed and may be confused in places • may include a relevant diagram to support their explanation.	4–7
1	• is very brief and/or lacks coherence • shows some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely • demonstrates very limited ability to apply relevant economic principles and/or data to the question • may include some very limited analysis but the analysis lacks focus and/or becomes confused • may include a diagram but the diagram is likely to be inaccurate in some respects or is inappropriate.	1–3

Relevant issues include:

- Recognition that a rise in fares can result from a fall in supply and/or a rise in demand
- Rising input costs — energy/diesel, electricity, wages following industrial action — shifting the supply curve to the left
- Index-linked regulated fare caps (RPI uplift) automatically raise headline fares when general inflation is high (Extract E)
- Falling subsidies — if the government reduces the £14bn+ taxpayer contribution (Extract E), TOCs must recover more of their costs from fares
- Capacity constraints — the high fixed cost natural-monopoly infrastructure (Extract E) means demand spikes (peak commuting) push fares up rather than expanding supply quickly
- Inelastic demand for many commuter journeys — TOCs can pass on cost rises with little loss of revenue
- Price discrimination — high Anytime fares relative to Advance fares (Extract F) raise the average fare; information failure means many consumers buy the more expensive option
- Rising demand from population growth, modal shift from car/air for environmental reasons
- Use of a relevant supply–demand diagram showing the shift(s) and the resulting rise in equilibrium fare
- Significance of XED with cars, coaches and air travel; significance of YED on long-distance leisure travel

MAXIMUM FOR QUESTION 31: 10 MARKS

3 2 Discuss whether the UK government should increase its subsidies to the rail industry.

[25 marks]

Areas for discussion include:

- Rail as a service generating positive externalities of consumption — reduced road congestion, lower CO■ emissions per passenger kilometre, better access to jobs and education (Extract E)

• Diagram of a positive externality of consumption showing MPB, MSB and the welfare loss from under-consumption; how a subsidy moves output toward the social optimum
• Rail as a natural monopoly — high fixed costs, very low marginal costs (Extract E); subsidies can support pricing closer to marginal cost than average cost (allocative efficiency vs sustainability)
• Modal shift from car and air travel as an explicit policy aim, in line with net-zero commitments
• Effects on different agents — commuters (Extract F notes mainly higher-income, South East), leisure travellers, freight customers, rail workers, motorists, taxpayers, the environment
• Equity concerns — Extract F argues subsidies benefit higher-income commuters while taxpayers across the country pay; case for targeting subsidies (e.g. regional, off-peak, low-income discounts)
• Financial cost of higher subsidies (already £14bn+ per year — Extract E) and the opportunity cost — NHS, schools, road maintenance, other priorities
• Alternatives to higher subsidies — simplification of ticketing (Extract F), public-sector operator reform, productivity gains, RPI-X style regulation to drive cost reduction
• Risks of government failure — over-investment in routes with low passenger demand, X-inefficiency in a subsidised public-sector operator, regulatory capture, complexity of franchise/operator arrangements
• Effect of changing patterns of demand — home working reduces peak demand (Extract E), which may weaken the case for subsidising additional peak capacity
• Case for leaving more to the market — competitive pressure on TOCs, ticketing innovation by third-party retailers
• Market failure versus government failure
• Equity versus efficiency
• An overall, supported judgement on whether subsidies should rise, fall or be redirected

The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the candidate's response.

Use the levels mark scheme on the earlier page to award marks for this question.

MAXIMUM FOR QUESTION 32: 25 MARKS